

Lab 7: Grayscale Stretch

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Two CUDA reductions find the minimum and maximum. A 2D map computes $(p - \min)255/(\max - \min)$. Constant images are preserved to avoid division by zero. Reduction time and map time for several blocks are reported separately. Reduction timing includes returning its two scalar results to the host. The extrema and output are checked against NumPy.

Kernel timing uses `time.time()` after JIT warm-up and CUDA synchronization, averaged over three GPU runs. Device allocation and host/device copies are excluded. CPU measurements are single runs. Speedup is relative to the stated CPU implementation. Actual measurements appear after running the notebook on Colab; no timings are assumed here.